

these neurons are found in the ventral pallidum of lizards and turtles. If confirmed, this would suggest that the dorsal pallidum in birds may have recruited gene regulatory programs that were ancestrally restricted to cell types in the ventral pallidum.

Evolutionary change in the brain can occur at every level of organization, from cell types to neural circuits and brain regions. Zaremba *et al.*, Hecker *et al.*, and Rueda-Alaíña *et al.* illustrate how comparisons across multiple levels are critical to understand brain evolution. Single-cell profiling is a powerful tool for reconstructing the evolution of neuron types while also characterizing brain regions. However, cell type comparisons may be insufficient to clarify the evolution of brain regions, particularly without developmental context. This is because the same cell identity program—the gene expression program that establishes and maintains type-specific characteristics (13)—can be activated in different developmental contexts. Astrocytes, for example, are the same cell type regardless of whether they develop from forebrain or hindbrain progenitors. In the chicken pallidum, ignoring the transcriptomic convergence of mixed-origin neuron types during development would lead to the erroneous conclusion that the bird dorsal and ventral pallidia are homologous to the lizard ventral pallidum. Developmental analyses indicate that distinct dorsal and ventral pallidia already existed in the last common ancestor of birds, lizards, and mammals.

Extending beyond the molecular comparison of cell types is even more critical to infer the evolution of neural circuits. As Rueda-Alaíña *et al.* show, nonhomologous neuron types can be part of neural circuits with remarkably similar architecture across regions and species. Conversely, Hecker *et al.* and Zaremba *et al.* identified neuron types in birds and mammals with similar transcriptomes but diverging connectivity. Integrating cell type molecular and connectivity data to resolve the evolution of neural circuits is one of the biggest challenges for future research. ■

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NEUROSCIENCE

Understanding the desire for dessert

A neural circuit in mice mediates the preference for high-sugar food after a meal

By Sadaf Farooqi

Overeating is a major cause of obesity, type 2 diabetes, cardiovascular disease, and some cancers (1). Although in many societies, readily available, cheap foods promote overconsumption, the sensory properties of food also influence the amount and type of food eaten. Sensory-specific satiety (fullness) describes the decline in pleasantness of a food as it is eaten, relative to uneaten foods

that have different sensory qualities—for example, the desire for a sweet dessert after a meal (2). However, the neural mechanisms that mediate this preference for sweet food are incompletely understood. On page 750 of this issue, Minère *et al.* (3) report a neural circuit that regulates the preference for sweet food after the consumption of a standard meal in mice (*Mus musculus*). This circuit is formed by neurons that project from the hypothalamus to the paraventricular thalamus, a region involved in sensing pleasantness and driving motivated behaviors. This work advances understanding of the drive to eat sweet food when already full.

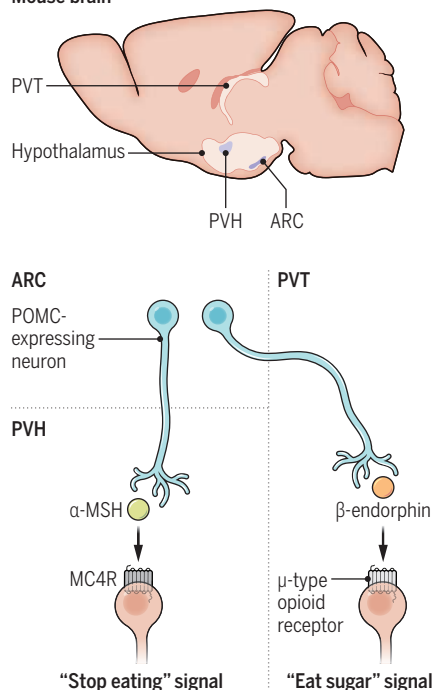
Eating behavior is modified by external environmental cues (visual appearance, smell, and availability of food) and internal drives, such as hunger and satiety (4). These external and internal cues are detected and integrated by highly conserved neural circuits, the functions of which are to defend against starvation and adapt to changing energy demands during growth and reproduction. Food also has rewarding properties that drive consumption. The desire for sweet foods is an innate behavior evident in multiple mammalian species; even newborn humans will consume a greater volume of a solution that tastes sweeter (5). In nonhuman species, naturally occurring plant sugars (for example, glucose, sucrose, fructose, and maltose) and lactose in milk induce the sensory quality of sweetness.

A more comprehensive understanding of the molecular mechanisms that underpin these highly conserved eating behaviors is now emerging. Neural circuits in the hypothalamus—a deep subcortical brain region surrounded by a relatively leaky blood-brain barrier—respond to changes in hormones and nutrients that circulate in the blood. A key circuit involves neurons in the arcuate nucleus of the hypothalamus that respond to the hormone leptin, which is released by adipose tissue and signals nutrient availability to the brain.

Signaling sweet preference

Pro-opiomelanocortin (POMC)-expressing neurons are activated in the satiated (full) state and project from the arcuate nucleus of the hypothalamus (ARC) to the paraventricular hypothalamus (PVH) and the paraventricular thalamus (PVT). In the PVH of mice, these neurons release α -melanocyte-stimulating hormone (α -MSH) that binds to the melanocortin 4 receptor (MC4R), reducing hunger and increasing satiety. POMC-expressing neurons that project to the PVT release β -endorphin, which binds to the μ -type opioid receptor, resulting in a desire for sugar.

Mouse brain



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These neurons express pro-opiomelanocortin (POMC) and connect to neurons in the paraventricular hypothalamus that express the melanocortin 4 receptor (MC4R). In the fed state, leptin stimulates the expression of POMC, which is processed into smaller peptides, including α -melanocyte-stimulating hormone (α -MSH) and β -endorphin. Binding of α -MSH to MC4R leads to reduced hunger, increased satiety, and a reduction in food intake. Conversely, β -endorphin activates δ -type and μ -type opioid receptors in limbic nuclei and other brain regions to give food a hedonic quality, and it can also affect the motivational properties, or wanting, of food (6).

Minère *et al.* investigated the role of β -endorphin in the response to fullness in mice. They found that β -endorphin-positive POMC-expressing neurons project to the paraventricular thalamus (PVT) (see the figure). Optogenetic stimulation of these neurons evoked β -endorphin release in the PVT. To understand whether this circuit has a role in the desire for sweet foods when full, the authors offered mice either a high-sugar diet or standard chow (with comparatively low sugar content) after a 90-min bout of feeding. In control conditions, mice offered the high-sugar diet ate more than those offered the chow. However, optogenetic inhibition of POMC neuron terminals in the PVT reduced the consumption of the high-sugar diet, indicating that this behavioral response was mediated by the POMC-PVT circuit.

Minère *et al.* suggest that therapies that reduce the preference for sweet food might be effective in the treatment of obesity. However, it is important to consider that the feeling of pleasure relating to food is closely linked to mood. Anhedonia (the reduced ability to experience pleasure) is a cardinal sign of depression, and sucrose preference tests are used as measures of anhedonia, depressive behavior, and the response to antidepressant treatment in mice and humans. Clinical trials of the antiobesity medication rimonabant (a cannabinoid 1 receptor antagonist targeting neural circuits involved in reward) showed significant weight loss. However, once the drug was licensed and larger numbers of people with obesity were treated, the risks of severe depression and suicidal ideation became evident, leading to its withdrawal from clinical use. These findings suggest that although medications that suppress sucrose preference might reduce total food intake, the ability to sense some pleasure from food is important for health.

"The desire for sweet foods is an innate behavior evident in multiple mammalian species..."

Understanding food preference in humans is further complicated by the observation that sweet foods are often also high in fat. Genetic disruption and pharmacological inhibition of melanocortin signaling in mice as well as MC4R deficiency in humans are associated with reduced desire for, and consumption of, high-sucrose solutions (mice) and food (humans) but an increased preference for high-fat diets (7, 8). Preference for foods high in fat might in part be mediated by MC4R-expressing neurons in the amygdala because the inhibition of MC4R in this brain region increases fat consumption in mice. Neurons in the central nucleus of the amygdala project to the parabrachial nucleus, which responds to gustatory and visceral signals and is involved in sucrose preference (9). Cumulatively, these findings and the data from Minère *et al.* suggest that interactions between melanocortin and opioid receptor-mediated signaling have a critical role in food preference.

A comprehensive understanding of the neural mechanisms involved in mediating the preference for dietary fat, sucrose, and other nutrients is needed to understand the drivers of overconsumption—a major cause of obesity and metabolic disease. Further studies are required to evaluate the relative contribution of sweet taste and type of sugar or sweeteners to liking, wanting, food seeking, and consumption, particularly when combined with fat. Understanding how neural circuits are affected by food intake after a single meal and after repeated consumption of a particular food (habituation) will also be important. Ultimately, policies aimed at improving the quality of food and tackling its adverse effects on health should be informed by an awareness of the powerful biological drives that shape sweet preference, sugar consumption, and eating behavior. ■

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HYDROGEN PRODUCTION

Thriftig iridium for hydrogen

Anchoring catalysts on an engineered oxide support enables stable water electrolysis

By Andrew D. Pendergast^{1,2,3} and Shannon W. Boettcher^{1,2,3,4}

Using renewable electricity to produce hydrogen fuel reduces reliance on fossil fuels. Proton exchange membrane water electrolyzers (PEMWEs) are the highest-performing commercialized technology (1). These devices split water into oxygen gas and hydrogen ions (protons) at the anode. The protons then migrate through an ion-conducting polymer membrane (ionomer) to be reduced to hydrogen gas at the cathode. The anode reaction's harsh environment requires the use of precious-metal catalysts, such as iridium oxide (IrO_x). Given the expense and scarcity, the design of electrodes that minimize the use of precious metals without compromising the requisite stability and activity is desired for large-scale hydrogen production (2, 3). On page 791 of this issue, Shi *et al.* (4) report that anchoring IrO_x catalysts onto porous cerium-oxide (CeO_x) supports maintains performance even with much reduced precious metal use.

Although many materials have been explored (5), IrO_x remains the only practical catalyst for oxygen evolution in PEMWEs because of its stability under acidic oxidizing conditions. Known degradation mechanisms include catalyst dissolution (2, 3), detachment, agglomeration, Ostwald ripening (dissolution and redeposition of small particles into large particles), active-site blocking by oxygen bubbles, and electrical or ionic isolation of catalyst particles (see the figure) (1, 6). To circumvent these issues, commercial PEMWE anodes are loaded with a high mass of IrO_x (2 to 3 mg cm^{-2}), despite the cost and scarcity of iridium (1).

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